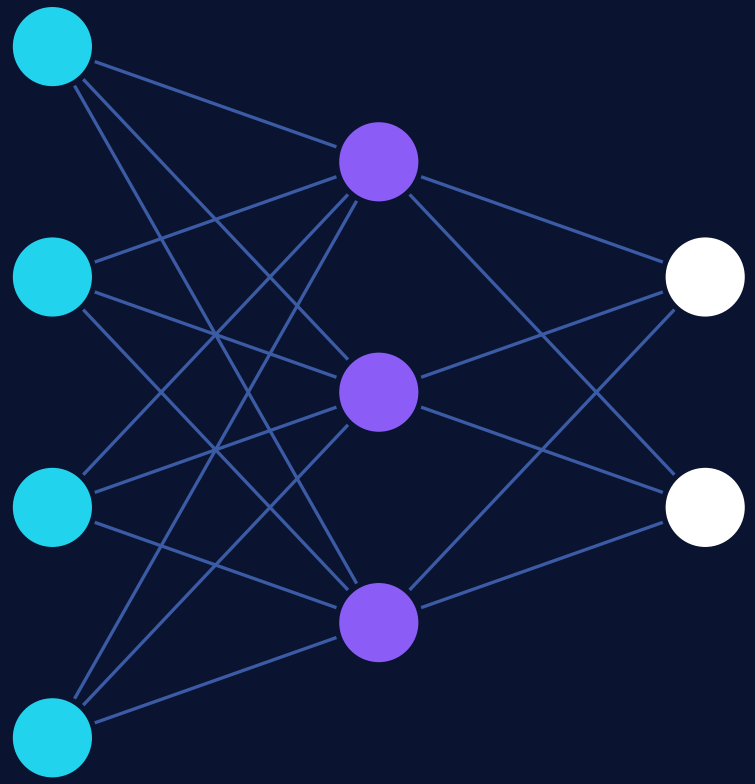
Machine Learning in Real Life

How Machines Learn from Data and Experience



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What Is Machine Learning?

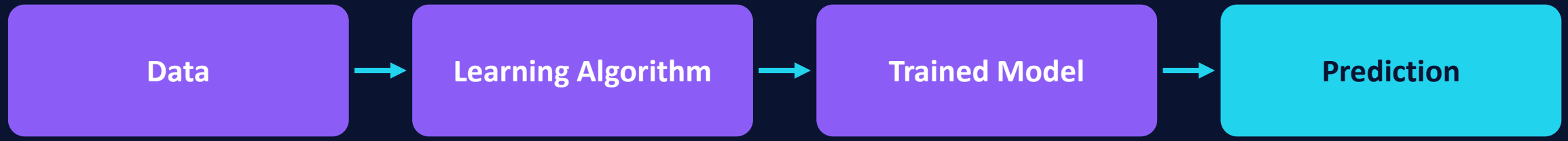
Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and improve their performance without being explicitly programmed for every task.

Key Points

- Learns patterns from data or experience
- Makes predictions or decisions
- Improves through training or feedback, depending on the method

Real-Life Examples

- Email spam detection
- Movie and video recommendations
- Face recognition on smartphones



Three Types of Machine Learning



Supervised Learning

- Learns from labelled examples
- Used for classification and prediction

Example: detecting spam emails or predicting house prices



Unsupervised Learning

- Finds patterns in unlabelled data
- Groups similar items or finds hidden structure

Example: grouping customers by shopping behaviour

MAIN FOCUS



Reinforcement Learning

- Learns through actions and feedback
- Gets rewards or penalties from an environment

Example: training a game-playing agent or a robot

Reinforcement Learning: Learning Through Experience

Reinforcement Learning (RL) is a type of Machine Learning in which an agent learns to choose actions by interacting with an environment and receiving rewards or penalties.

1. The agent observes the current situation
2. It selects an action
3. The environment responds
4. The agent receives a reward or penalty
5. It learns from the outcome and improves future decisions



Example: Robot navigating a room

- The robot is the agent; the room is the environment
- Moving forward or turning is an action
- Reaching the destination gives a positive reward
- Hitting an obstacle can give a negative reward

Rewards and penalties are feedback signals, not money or physical punishment.

Core Components of Reinforcement Learning

1 Agent

The learner or decision-maker that takes actions

Example: *A robot learning to navigate*

2 Environment

The world or system the agent interacts with

Example: *The room in which the robot moves*

3 State (s)

The current situation of the agent in the environment

Example: *The robot's position and nearby obstacles*

4 Action (a)

A possible move or choice the agent can make

Example: *Move forward, turn left, or turn right*

5 Reward (r)

Feedback showing how useful an action was

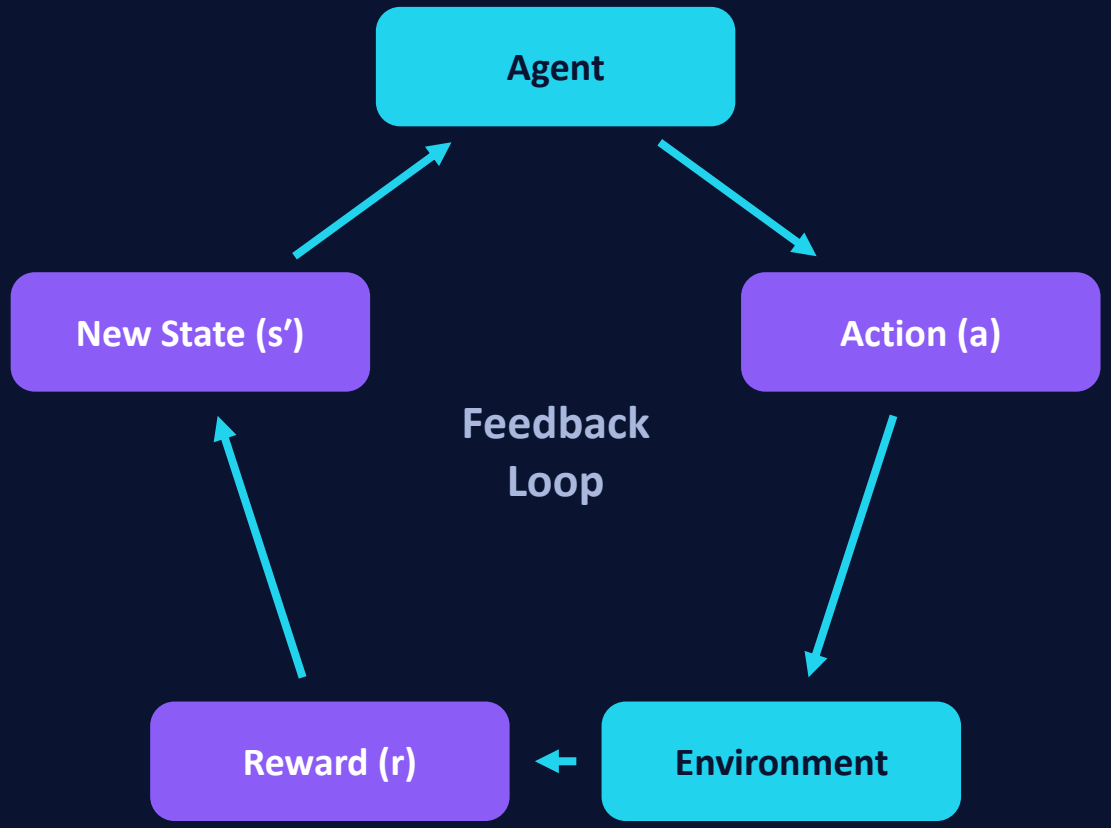
Example: *A positive reward for reaching the destination*

6 Policy (π)

The strategy the agent uses to choose an action in a state

Example: *Turn left when an obstacle blocks the path*

How It Works: The Feedback Loop



1. Agent observes the current state of the environment
2. Agent selects an action using its policy
3. The environment changes in response to the action
4. The environment gives a reward and the new state
5. The agent uses this experience to improve future decisions, then the loop returns to Step 1

The agent learns over many repeated interactions. Not every action gets an immediate reward, and learning is gradual, not perfect.

Example: A robot moves toward a target, receives positive feedback when it gets closer, and gradually learns a better route.

Real-Life Applications and Conclusion

Real-Life Applications



Gaming
Learning strategies to win games



Robotics
Learning movement and task execution



Autonomous systems
Supporting navigation and control



Recommendation and resource management
Sequential decisions where appropriate

Conclusion

- ML helps computers learn patterns and make predictions or decisions
- Supervised, Unsupervised, and Reinforcement Learning solve different problems
- RL uses an agent, environment, state, action, reward, and policy
- Feedback and repeated experience help an agent develop better strategies

Machine Learning turns data and experience into smarter decisions. Thank you!

Questions?